Yizun Yan

Department of Agricultural and Consumer Economics
University of Illinois Urbana-Champaign
yizuny2@illinois.edu | +1 (217) 721-9182 | yizunyan.com

EDUCATION

Ph.D. in Agricultural and Applied Economics, University of Illinois Urbana-Champaign — Expected 2027
M.S. in Environmental and Resource Economics, Renmin University of China — 2022
B.S. in Management, Northwest Agricultural & Forestry University, China — 2020

RESEARCH INTERESTS

Environmental and Resource Economics, Land Use, Climate Adaptation, Producer Decision-Making, Nonmarket Valuation

JOB MARKET PAPER

"Species Diversification for Climate Adaptation in Perennial Systems: Trading off Annual Costs for Long-term Resilience Benefits," *with Shadi Atallah.* [Link]

Abstract: Species diversification can enhance the resilience of agroecosystems to climate-related shocks and serve as a climate adaptation strategy. We study how producers in perennial systems weigh the annual costs against the delayed resilience benefits of species diversification, and how climate projection information shifts this trade-off. Given that private investments in resilience generate public good benefits, we test whether producers view diversification by others as strategic complements or substitutes. We develop a dynamic optimization model of a producer choosing the optimal diversification intensity and test its predictions with U.S. maple producers using a discrete choice experiment with a randomized climate-impact information treatment. The annual payment required rises by about $47 per acre for each percentage point increase in diversification intensity. Older producers require more than twice as much payment as their younger peers when asked to commit to longer contracts, and most prefer to keep their current monocultures. Through a salience effect, region-specific climate projection information shifts how producers weigh costs and benefits over time, by increasing the weight they place on future resilience benefits and reducing the weight placed on annual yield losses. Peer adoption shifts this weighing across space, as producers view diversification by peers as a strategic complement to their own investments, reducing the required payment by 29% when 50% of peers have diversified. These results suggest three strategies for conservation programs seeking to increase agroecosystem resilience through species diversification while lowering costs: accounting for older producers' temporal constraints in longer contracts, coordinating adoption among neighbors, and pairing payments with climate information.

WORKING PAPERS

"Diversification as a Portfolio Decision: Evidence from a Choice Experiment in the U.S. Corn Belt," *with Shadi Atallah and Ben Gramig.*

Abstract: This paper studies how farmers evaluate agricultural diversification alternatives that differ in expected returns, income risk, and production horizons. Using a choice experiment with producers in the U.S. Corn Belt, we examine how these economic considerations affect both the adoption and scale of diversification. The analysis sheds light on the relative importance of barriers to diversified production and informs the design of policies intended to address them.

"New England Maple Producers May Be Open to Diversifying the Species Composition of Their Sugarbushes," *with Shadi Atallah, Muhammad Jawad Khan, and David Moore.*

Abstract: Maple syrup production in New England is projected to decline by about 50% by 2100 under climate change, and diversifying maple forests can enhance their resilience to extreme weather. We use a choice experiment with producers recruited at a maple industry conference in Massachusetts to elicit preferences for sugarbush diversification programs, focusing on diversification intensity given a trade-off between long-run resilience benefits and immediate

yield losses. Producers favored moderate levels, possibly because of perceived diminishing marginal benefits of diversification due to interspecies competition. They also favor longer contracts, consistent with the slow maturation of resilience benefits.

"Trade Liberalization and Urban Spatial Structure: Evidence from China's WTO Accession," *with Shang Xu and Hongliang Zhang.*

Abstract: We analyze how international trade shapes urban spatial structure, exploiting differential exposure to trade liberalization following China's accession to the World Trade Organization. Our findings show that trade liberalization accounted for 38% of China's urban expansion between 1995 and 2005. It also made cities more compact, reducing the average within-city distance by 8.9 kilometers. These effects are largely driven by indirect export exposure through industry input-output linkages, which increase population and firm density, boost household savings, and expand financial institution lending.

WORK IN PROGRESS

"Valuing Soil Microbiome-Mediated Ecosystem Services in Farm Management Decisions," *with Shadi Atallah, Elias Bloom, and Clare Casteel.*

Abstract: This study examines how farmers value ecosystem services provided by soil microbiomes and how information about these otherwise difficult-to-observe services affects farm management decisions. Combining soil sampling and laboratory analysis on New York farms with a choice experiment, we link measured microbiome performance to farmers' valuation and adoption of microbiome-supportive practices. The study provides a producer-side valuation of soil natural capital and examines how peer rankings of microbiome performance shape its management.

EXTENSION

"An Operation-Specific Decision Tool for Species Diversification in Sugarbush" 2026
Finalist, AAEA Extension Competition, Kansas City, MO

Abstract: This interactive decision tool helps maple syrup producers assess whether species diversification in their sugarbush fits their operation and, if so, at what level of diversification intensity. It combines financial cost-benefit analysis with producer preferences estimated from a choice experiment, starting from survey-based default values that producers replace with their own operation, market, and planning-horizon information. The tool supports both self-guided use and consultations with extension educators and foresters.

CONFERENCE AND SEMINAR PRESENTATIONS

"Species Diversification as Climate Adaptation in Perennial Systems: Trading off Annual Costs for Long-term Resilience Benefits"

Heartland Environmental and Resource Economics Workshop, Urbana, IL 2026

"Diversification as a Portfolio Decision: Evidence from a Choice Experiment in the U.S. Corn Belt"

Agricultural and Applied Economics Association (AAEA) Annual Meeting, Kansas City, MO (poster) 2026

"Maple Syrup Producers' Willingness to Diversify Forests for Increased Resilience"

JEEM/REE Workshop on Nonmarket Valuation, Bilbao, Spain 2026
7th World Congress of Environmental and Resource Economists (WCERE), Carcavelos, Portugal 2026
Program in Environmental and Resource Economics (pERE), Urbana, IL 2025
XXVI Annual BIOECON Conference, Cambridge, UK 2025
Association of Environmental and Resource Economists (AERE) Conference, Santa Ana Pueblo, NM 2025
Agricultural and Applied Economics Association (AAEA) Annual Meeting, New Orleans, LA 2024
W5133 Multistate Research Project Meeting, Fort Collins, CO 2024

"International Trade and Urbanization: Evidence from China"

16th North American Meeting of the Urban Economics Association, Washington, D.C. 2022

RESEARCH EXPERIENCE

Research Assistant (Prof. Shadi Atallah), University of Illinois Urbana-Champaign 2023–Present

- Contributed to annual progress reports for two USDA-NIFA projects.
- Coordinated a multi-state producer survey with the UNH Survey Center, including building a contact database of 1,483 maple syrup producers.
- Mentored an M.S. student in choice model estimation and supervised two research assistants.

Research Assistant (Prof. Shang Xu and Prof. Hongliang Zhang), Renmin University of China 2020–2022

- Constructed a prefecture-level panel linking customs, employment, input-output, satellite land-use, and historical road network data.

Research Assistant, Chinese Academy of Agricultural Sciences 2020

- Reviewed research on payments for ecosystem services and ecological compensation mechanisms.

TEACHING EXPERIENCE AND TRAINING

Instructor of Record, University of Illinois Urbana-Champaign 2026–Present
ACE 161: Microcomputer Applications

Participant, 3rd Annual AAEA Teaching Academy, Kansas City, MO 2026

Teaching Assistant, University of Illinois Urbana-Champaign 2023–2024
ACE 474: Economics of Consumption; ACE 476: Behavioral Economics

Teaching Assistant, Renmin University of China 2021–2022
Intermediate Microeconomics; Development Economics

PRODUCER ENGAGEMENT

Agroforestry for Conservation and Recreation Workshop (information booth), Monticello, IL 2025
Farmer Focus Group for Corn Belt Diversification Survey (facilitator, 4 farmers), Urbana, IL 2025
Backyard Maple Syrup Workshop (information booth), Dixon Springs, IL 2025
International Maple Syrup Conference (information booth), Sturbridge, MA 2023

SERVICE AND LEADERSHIP

Reviewer, *PLOS ONE* 2026
Student Coordinator, Program in Environmental and Resource Economics (pERE) Seminar 2025
Student Coordinator, Heartland Workshop (UIUC) 2025
Graduate Student Organization Officer 2024–2025

HONORS AND AWARDS

Finalist, AAEA Extension Competition 2026
Agricultural and Applied Economics Association (AAEA) Trust Travel Grant 2026
ACE Travel Award, University of Illinois Urbana-Champaign 2025, 2026
Dunn & Linse Scholarship, University of Illinois Urbana-Champaign 2023
Graduate Student Fellowship, Renmin University of China 2021, 2022
China National Scholarship, Ministry of Education of China 2019

SKILLS

Programming: Stata, R, Python
Software & Tools: Ngene, ArcGIS, Qualtrics, GAUSS, JMP, LaTeX
Languages: Chinese (Native); English (Fluent)

REFERENCES

Shadi S. Atallah
Associate Professor
Department of Agricultural and Consumer Economics
Associate Director, Center for the Economics of Sustainability
University of Illinois Urbana-Champaign
satallah@illinois.edu

Madhu Khanna
ACES Distinguished Professor in Environmental Economics
Director, Institute for Sustainability, Energy and Environment
Director, Center for the Economics of Sustainability
University of Illinois Urbana-Champaign
khanna1@illinois.edu

Ben Gramig
Associate Professor
Charles H. Dyson School of Applied Economics and Management
College of Agriculture and Life Sciences
Cornell University
bmg99@cornell.edu

Mo Zhou
Associate Professor of Forest Economics and Management
Department of Forestry and Natural Resources
Purdue University
mozhou@purdue.edu

WORK AUTHORIZATION

Citizenship: China
Work Authorization: F-1 visa; eligible for up to 3 years of STEM OPT